

Neil **Lu**

Astrophysicist

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About Me

I work in gravitational wave astrophysics, analyzing data from binary black-hole mergers to test general relativity and probe black hole physics. I am a member of the LIGO-Virgo-KAGRA collaboration and have contributed as an data-analyst and on the paper writing team to several full-collaboration papers. Outside of research I am passionate about science outreach, promoting inclusion, and mental health in academic spaces.

Education

2023 – 2026 **Doctorate of Philosophy**
Australian National University
Primary supervisor: Ling Sun. Supervisory panel: Karl Wette, and Susan Scott
Thesis: Probing gravity and fundamental physics from binary black holes

Apr 2023 – June 2024 **Visiting Scholar**
Sapienza University of Rome
Supervised by Cristiano Palomba

Mar 2023 **Visiting Scholar**
Caltech
Collaborated with Sizheng Ma, and Yanbei Chen

2018 – 2021 **Bachelor of Philosophy (Science)**
Australian National University
1st class honours
Honours Thesis: Inferring neutron star physics with a continuous wave detection
Supervisors: Karl Wette, and Susan Scott

Publications

My publications are also available at this link.

First-Author

1. **Lu, N.**, Palomba, C., Piccinni, O. J., Sun, L. 2026. *A semicoherent resampling method for long-transient gravitational wave searches with applications to subsolar-mass primordial black-hole binaries* arXiv: 2609.12476. Submitted to Classical and Quantum Gravity.
2. **Lu, N.**, Ma, S., Piccinni, O. J., Chen, Y., Sun, L. 2026. *GW250114 Reveals Signatures of Post-Merger Black-Hole Horizon* Nature 655 (8121): 71–75.
3. **Lu, N.**, Ma, S., Piccinni, O. J., Sun, L., Finch, E. 2025. *Statistical Identification of Ringdown Modes with Rational Filters.* Physical Review D 112 (6): 064047.
4. **Lu, N.**, Wette, K., Scott, S. M., Melatos, A. 2023. *Inferring Neutron Star Properties with Continuous Gravitational Waves.* Monthly Notices of the Royal Astronomical Society 521 (2): 2103–13.

Co-Author

I have been a member of the LIGO Scientific Collaboration since 2023 and am a co-author on over 50 full-collaboration papers since April 2024. Included below are those to which I have made a direct personal contribution.

1. Nathan, R. S., Hido, S., **Lu, N.**, et al. (2026). *Accounting for detector calibration uncertainty in black-hole ringdown analysis*. arXiv:2610.11052. Submitted to Physical Review D.
2. Weller, C., Laeuger, A., Li, F., et al. (including **Lu, N.**) 2026. *From driven oscillations to free ringdown: a particle plunging into Kerr* arXiv:2609.09531. Submitted to Physical Review Letters.

3. Sun, L., Ma, S., Chen, P. P., et al. (including **Lu, N.**) 2026. *Complex frequency evolution of direct waves from binary black hole mergers* arXiv:2608.23209. Submitted to Physical Review D.
4. Sun, L., K. Kuns, Slagmolen, B. J. J., et al. (including **Lu, N.**) 2026. *LIGO A$^\sharp$: Detector Design and Science Prospects Beyond A+.* arXiv:2608.11673. Submitted to Classical and Quantum Gravity.
5. The LIGO-Virgo-KAGRA Collaboration (including **Lu, N.**) . 2026. *GWTC-4.0: Tests of General Relativity. III. Tests of the Remnants.* arXiv:2603.19021. Submitted to Astrophysical Journal Letters.
6. The LIGO-Virgo-KAGRA Collaboration (including **Lu, N.**) . 2026. *GWTC-5.0: Tests of General Relativity.* arXiv:2607.19293. Submitted to Astrophysical Journal Letters.
7. The LIGO-Virgo-KAGRA Collaboration (including **Lu, N.**) . 2025. *Black Hole Spectroscopy and Tests of General Relativity with GW250114.* Physical Review Letters 136 (4): 041403.
8. The LIGO-Virgo-KAGRA Collaboration (including **Lu, N.**) . 2025. *GW230814: Investigation of a Loud Gravitational-Wave Signal Observed with a Single Detector.* The Astrophysical Journal Letters 1004 (2): L23. Member of the paper writing team

Presentations and Conferences

1. MIT GRITTS seminar - *invited talk* — Cambridge, United States, 2026
2. Perimeter Institute Strong Gravity seminar - *invited talk* — Waterloo, Canada, 2026
3. American Physical Society Global Physics Summit - *contributed talk* — Denver, United States, 2026
4. OzGrav-KAGRA early-career researcher school - *science organising committee* — Kanazawa, Japan, 2026
5. Australian Institute of Physics Summer Meeting - *contributed talk* — Wollongong, Australia, 2025
6. OzGrav early career researcher workshop - *science organising committee* — Melbourne, Australia, 2025
7. Edoardo Amaldi Conference on Gravitational Waves - *contributed poster* — Glasgow, United Kingdom, 2025
8. LIGO-Virgo-KAGRA meeting - *invited talk* — Melbourne, Australia, 2025
9. Gravitational Wave Physics and Astronomy Workshop - *contributed poster* — Birmingham, United Kingdom, 2024
10. Virgo week - *contributed talk* — online, 2024
11. Gravity Shape Pisa - *contributed talk* — Pisa, Italy, 2023
12. Future Perspectives on primordial black holes - *contributed talk* — Rome, Italy, 2023
13. Edoardo Amaldi Conference on Gravitational Waves - *contributed talk* — online, 2023
14. LIGO-Virgo-KAGRA meeting - *contributed talk* — online, 2023

Awards, Outreach, Service

1. **2026** John Carver Prize, ANU
2. **2026** Three Minute Thesis, ANU College of Science, 1st place
3. **2026** Press releases "A 'direct wave' from colliding black holes reveals signature of a whirlpool in spacetime "Scientists Find a Way to Study the Event Horizon Where Light & Sound Are Swallowed for Eternity", "Messages from the surface of a black hole"
4. **2024-2026** Center for Gravitational Astrophysics, student representative
5. **2024-2026** OzGrav early-career researcher committee
6. **2025** OzGrav rising stars, honorable mention
7. **2025** Press release Gravitational Waves Confirm Hawking's Prediction
8. **2024** Three minute talks, OzGrav meeting, 1st place